

Week 8 Final Integration Readiness & HC-POD Documentation

HealthConnect Clinic – Data Science Track

AnalystLab Africa Experience Lab, Week 8

Part 1 — Final Integration Readiness

Requirement	Response
What is my final component?	A Random Forest model predicting whether a HealthConnect appointment will end in a no-show, plus the feature-engineering logic that turns a raw appointment record into model-ready inputs.
What problem does it address?	HealthConnect misses a large share of scheduled appointments; the model flags which appointments are at higher risk so staff can prioritise outreach.
What is its current status?	Tested and validated (Week 7): no overfitting, stable across 5-fold grouped cross-validation (ROC-AUC 0.677 ± 0.008), and shown to be useful at a realistic top-20%-risk threshold (0.722 precision vs. 0.500 base rate). No further model changes made in Week 8 — this week finalises documentation and the ML Engineering handoff.
What did I improve during Week 7?	Confirmed the Week 6 candidate's stability through cross-validation and found and documented a pipeline input-validation gap (unseen categories silently absorbed), rather than changing the model itself, since testing found no weakness in the model that justified refinement.
Which other track(s) does it depend on?	Data Analytics (their segment-level findings, when available, could sharpen feature choices) and Machine Learning Engineering (needs this model and its input/output contract to build the production pipeline).
Which track(s) depend on my output?	Machine Learning Engineering depends on the final model and its documented input schema/output format to integrate it into the HealthConnect workflow.
What output will I provide?	The final trained model's expected input schema (11 raw fields + 4 engineered features), its output format (a 0–1 risk probability), and the known unseen-category handling issue that needs to be addressed at the pipeline level.
What output do I need from another track?	From ML Engineering: confirmation of how the pipeline will validate categorical inputs, so the unseen-category gap identified in Week 7 is closed before any production use.
What evidence demonstrates readiness?	Week 6 error analysis and model comparison, Week 7 overfitting/cross-validation/business-relevance testing, and this week's final model documentation notebook (week8_final_model_documentation.ipynb) and one-page summary.
What limitations remain?	Performance plateaus around 0.68 ROC-AUC regardless of algorithm tried; the model is a prioritisation tool, not suitable for automated decisions; the pipeline-level unseen-category gap is documented but not yet fixed (ML Engineering scope).

Mandatory HC-POD Final Integration

Track collaborated with. Machine Learning Engineering, attempted. No ML Engineering pipeline was available within the HC-POD to test or integrate against by the time of this submission.

Dependency. Per the HealthConnect final integration matrix, the final Data Science model and its technical requirements are meant to support the final ML Engineering workflow.

Output received. None — no ML Engineering artefact or pipeline specification was available to receive.

Output provided. The final model's input schema (11 raw fields plus 4 engineered features), its output format (a risk probability rather than a hard classification), a recommended decision approach

(risk ranking, e.g. top 20%, rather than the default 0.5 cutoff), and the documented unseen-category handling gap identified during Week 7 testing.

Final integration activity. In the absence of an ML Engineering counterpart, the feature-building function itself was packaged and documented (Section 6 of `week8_final_model_documentation.ipynb`) as the reference implementation, including its known limitation, so that whoever integrates the pipeline can pick it up directly.

What changed. A concrete, testable handoff package now exists (input schema, output contract, known issue) where previously only informal notes existed across the Week 6–7 notebooks.

How the change improved the solution. Reduces the risk of late-stage integration surprises in a hypothetical production stage, by surfacing the unseen-category issue and the recommended decision threshold before any pipeline is built around the model.

Evidence. Section 6 ("Cross-Track Contribution to ML Engineering") of `week8_final_model_documentation.ipynb`; this document; the Week 7 notebook's pipeline robustness test that originally surfaced the issue.

Contribution to the final presentation. This handoff package is the basis for explaining, in the final video, what the model needs from ML Engineering and why the unseen-category issue matters for a real deployment.

As documented above and in the Week 6/7 submissions, no two-way exchange with another track's actual output occurred this cohort — Data Analytics did not respond in Week 6, and no ML Engineering pipeline was available in Week 7–8. This is reported transparently rather than presented as a completed two-way collaboration.